

Quiz 1 — Grading Rubric

MA4.101 Real Analysis · Monsoon 2026 · IIIT Hyderabad

Total marks: 15 + 2 bonus (Q2d)

General Points

The rubric provided here is a general guide for how marks are given for questions.

Alternate approaches to questions are also considered and marked accordingly as long as it is mathematically correct and rigorous.

Even if an approach is not completely correct, partial marking is also awarded based on the idea and meaningful execution, on the discretion of the correcting TA.

However, completely meaningless answers will not be awarded any marks.

You get credit for having the right mathematical idea, not for how it is presented. In particular, you will **not** lose marks for:

- not naming axioms explicitly, as long as it is mentioned explicitly or implicitly
- skipping trivial steps
- stating results a little informally

You **will** lose marks if:

- a key case is missing entirely;
- a claim is mathematically wrong (not just imprecise);
- the argument is circular (it assumes what it is trying to prove);
- an unproven fact is used without any acknowledgement.

Question 1 (6 marks)

A. Auxiliary inequality: $2n + 1 \leq n^2$ for all $n \geq 3$ — 2 marks

Checkpoint	Marks
Auxiliary claim stated, even implicitly as a step inside the main proof	0.25
Base case verified at $n = 3$ (or $n = 4$, see notes)	0.50
Inductive step: writes $2(n+1)+1$ in terms of $2n+1$ and applies the IH	0.75
Completes the step: bounds the expression by $(n+1)^2$	0.50
Subtotal	2

Notes:

- Using $2n+1 \leq n^2$ inside the main proof without a separate proof still earns 1 mark here, and is not penalised in Part B.
- Proving it for $n \geq 4$ instead of $n \geq 3$ is fine, as long as the main induction still holds together.
- A combined argument, where this inequality is briefly justified inside the main proof, can earn close to full marks across A and B.

B. Main induction: $n^2 \leq 2^n$ for all $n \geq 4$ — 2 marks

Checkpoint	Marks
Proof by induction clearly set up (base + step, even informally)	0.25
Base case $n = 4$ correctly verified: $4^2 = 16 \leq 16 = 2^4$	0.50
Inductive hypothesis stated	0.25
Key step: $(n+1)^2$ bounded above by $2n^2$ using the auxiliary result	0.50
Applies the IH to get $2n^2 \leq 2 \cdot 2^n = 2^{n+1}$ and concludes	0.50
Subtotal	2

Notes:

- Writing $2n^2 \leq 2 \cdot 2^n = 2^{n+1}$ without explicitly naming the IH still earns the mark.
- Minor algebra slips that don't affect the structure of the argument are not penalised.

C. Full range determination — 2 marks

Checkpoint	Marks
Shows $n = 3$ fails: $3^2 = 9 > 8 = 2^3$ (a brief check suffices)	0.50
Checks at least two of $n = 0, 1, 2$ directly	0.50
States the complete correct answer: all $n \in \mathbb{N}$ except $n = 3$, i.e. $\{0, 1, 2\} \cup \{n \in \mathbb{N} : n \geq 4\}$	1.00
Subtotal	2

Notes:

- Correct final answer but only $n = 3$ checked (not $0, 1, 2$): 1.5 marks.
- All three small cases checked, but the final set not stated explicitly: 1.5 marks.
- Informal phrasing such as “fails only at $n = 3$ ” counts as a correct final answer.

Question 2 (9 marks + 2 bonus)

(a) Uniqueness of representation — 2 marks

Goal: show $f(a)+b = f(c)+d$ with $b \leq g(a)$, $d \leq g(c)$ implies $a = c$ and $b = d$.

Checkpoint	Marks
Sets up a contradiction from $a \neq c$ (WLOG or both cases, either is fine)	0.50
Derives the required contradiction using any valid combination of $b \leq g(a)$, $d \leq g(c)$, and (P1)/(P2); graded on soundness and extent	1.00
Concludes the contradiction, then deduces $b = d$	0.50
Subtotal	2

Notes:

- Working out only $a < c$ without saying “WLOG” is fine if correct — the other case is symmetric.
- Either (P1) or (P2) may be used. The 1-mark row is graded on how complete the inequality chain is: roughly 0.5 for the right idea with a real gap, 0.75 for a near-complete chain, 1.0 for a fully valid chain.
- $b = d$ via “subtract $f(a)$ from both sides” earns the mark.

(b) $S_{f,g} = \mathbb{N}$ when equality holds everywhere — 4 marks

Goal: prove $\mathbb{N} \subseteq S_{f,g}$ by induction, given $f(a+1) = f(a)+g(a)+1$ for all $a \in \mathbb{N}$.

Checkpoint	Marks
Induction strategy clearly indicated	0.50
Base case $n = 0$: writes $0 = f(0)+0 \in S_{f,g}$	0.50
IH: $n \in S_{f,g}$ correctly unpacked as $n = f(a)+b$ with $b \leq g(a)$	0.50
Case 1 ($b < g(a)$): $n+1 = f(a)+(b+1) \in S_{f,g}$	1.00
Case 2 ($b = g(a)$): uses the premise to write $n+1 = f(a+1)+0 \in S_{f,g}$	1.00
Conclusion: $\mathbb{N} \subseteq S_{f,g}$, hence $S_{f,g} = \mathbb{N}$	0.50
Subtotal	4

Notes:

- The bound $b \leq g(a)$ is necessary. If a case entirely omits checking it, that case earns 0.75 instead of 1 (Case 1 needs $b+1 \leq g(a)$; Case 2 needs $0 \leq g(a+1)$).
- Both cases argued together in one paragraph is fine
- Omitting “ $S_{f,g} \subseteq \mathbb{N}$ by definition” from the conclusion is fine

(c) $S_{f,g} \neq \mathbb{N}$ when equality fails somewhere — 3 marks

Goal: exhibit a specific $n_0 \in \mathbb{N}$ with $n_0 \notin S_{f,g}$.

Checkpoint	Marks
Genuine attempt to exhibit a counterexample (recognises that a specific n_0 must be produced, even if the choice is ultimately wrong)	0.50
Identifies a valid candidate $n_0 = f(a_0)+g(a_0)+1$ for some a_0 where equality fails	0.50
Case $a \leq a_0$: shows $f(a)+b < n_0$ (any valid approach; graded on soundness and extent — see also the alternative below)	1.00
Case $a > a_0$: shows $f(a)+b > n_0$ (any valid approach; graded on soundness and extent — see also the alternative below)	1.00
Subtotal	3

Notes:

- Using $f(a_0)+g(a_0)+2 \leq f(a_0+1)$ without deriving it costs nothing in the $a > a_0$ case.
- Partial credit is generous in both case rows: a directionally correct approach with the right ingredients earns most or all of the mark, even if a step is asserted rather than proved. One unified argument covering both directions earns full marks on both rows.
- Alternative (2 marks total): if the case-by-case argument isn't completed, you can instead earn 0.5 for recognising that $n_0 \notin S_{f,g}$ must be checked for *all* a and all valid $b \leq g(a)$, plus 0.25 for splitting this into the cases $a \leq a_0$ and $a > a_0$.
- Not stating $n_0 \in \mathbb{N}$ explicitly, or omitting the closing line, costs nothing.
- A reasonable but incorrect n_0 (e.g. off by one), used consistently: you earn the 0.50 for the attempt but not the 0.50 for a valid candidate; the case rows are graded on correct use of that candidate.

(d) Prove (P1) and (P2) from Eq. (1) — bonus, 2 marks

Attempt only after finishing (a)–(c). Partial credit is given generously.

Checkpoint	Marks
Shows $f(a) < f(a+1)$ from Eq. (1), using $g(a)+1 \geq 1$ for strict inequality	0.50
Proves (P2) by induction on $c \geq a+1$, with correct base (Eq. (1)) and step	0.75
Proves (P1) from (P2): $f(a) < f(a)+g(a)+1 \leq f(c)$ when $a < c$	0.75
Subtotal	2

Notes:

- (P1) proved directly by induction on c , rather than via (P2): full credit.
- A correct but slightly informal inductive step for (P2): full credit.
- Only one of (P1)/(P2) proved: up to 1 mark.